24/9/25

9. Design a Simple Topology and Configure One Router, Two Switches and PCs using Cisco Packet Tracer

Aim: To design and configure a simple network topology consisting of one router, two switches, and four PCs using Cisco Packet Tracer.

Procedure:

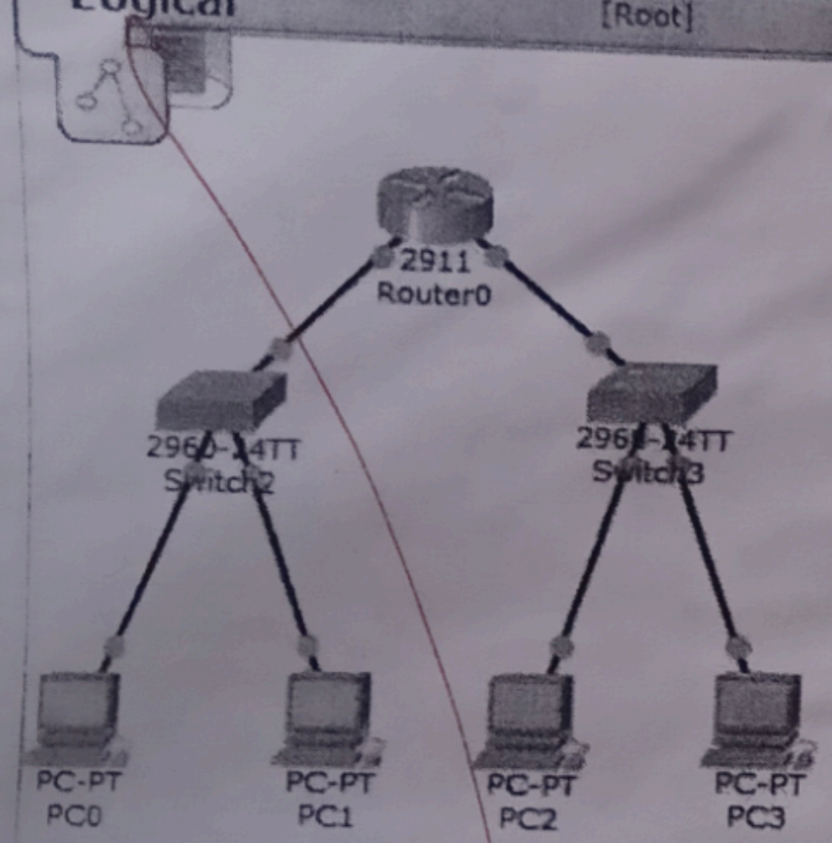
1. Open Cisco Packet Tracer.
2. Add Devices
3. Connect the Devices with the following connections
4. Open Router CLI
5. Configure Interface G0/0
6. Configure Interface G0/1
7. Save Router Configuration
8. Configure PC0
9. Configure PC1
10. Configure PC2
11. Configure PC3
12. Test Connectivity

Result: Designing a Simple Topology and configuring one router, Two switches and PCs using Cisco Packet Tracer has been executed successfully.

Cisco Packet Tracer Student

File Edit Options View Tools Extensions Help

Logical [Root]



Router0

Physical Config CLI

IOS Command Line Interface

```
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to up
exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

Copy Paste

PC0

Physical Config Desktop Custom Interface

Command Prompt

```
PC>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=0ms TTL=127
Reply from 192.168.2.2: bytes=32 time=0ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=4ms TTL=127
Reply from 192.168.2.2: bytes=32 time=0ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=0ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms

PC>
```

Time: 00:40:04 Power Cycle Devices Fast Forward Time

Realtime

Connections

Copper Straight-Through

Scenario 0 New Delete Toggle PDU List Window

Fire Last Status Source Destination Type Color Time(sec) Periodic